Maulana Abul Kalam Azad University of Technology, West Bengal

(Formerly West Bengal University of Technology)

Syllabus for B. Tech in Robotics

(Applicable from the academic session 2020-2021)

Curriculum Structure

First Year First Semester							
Mandatory Induction Program- 3 weeks duration							
Sl No.	**Category**	**Subject Code**	**Subject Name**	**Total Number of contact hours**			**Credits**
				L	**T**	**P**	
Theory							
1	Basic Science course	BS-PH101	Physics-I	3	1	0	4
2	Basic Science course	BS-M102	Mathematics –IB	3	1	0	4
3	Engineering Science Courses	ES-EE101	Basic Electrical Engineering	3	1	0	4
	Total Theory			**9**	**3**	**0**	**12**
Practical							
1	Basic Science course	BS-PH191	Physics-I Laboratory	0	0	3	1.5
2	Engineering Science Courses	ES-EE191	Basic Electrical Engineering Laboratory	0	0	2	1
3	Engineering Science Courses	ES-ME192	Workshop/ Manufacturing Practices	1	0	4	3
	Total Practical			**1**	**0**	**9**	**5.5**
			Total of First Semester	**10**	**3**	**9**	**17.5**
First Year Second Semester							
Sl No.	**Category**	**Subject Code**	**Subject Name**	**Total Number of contact hours**			**Credits**
				L	**T**	**P**	
Theory							
1	Basic Science course	BS-CH201	Chemistry-I (Gr-A)	3	1	0	4
2	Basic Science course	BS-M202	Mathematics –IIB	3	1	0	4
3	Engineering Science Courses	ES-CS201	Programming for Problem Solving	3	0	0	3
4	Humanities and Social Sciences including Management courses	HM-HU201	English	2	0	0	2
	Total Theory			**11**	**2**	**0**	**13**
Practical							
1	Basic Science course	BS-CH291	Chemistry-I Laboratory	0	0	3	1.5
2	Engineering Science Courses	ES-CS291	Programming for Problem Solving	0	0	4	2
3	Engineering Science Courses	ES-ME291	Engineering Graphics & Design (Gr-A)	1	0	4	3
4	Humanities and Social Sciences including Management courses	HM-HU291	Language Laboratory	0	0	2	1
	Total Practical			**1**	**0**	**13**	**7.5**
			Total of Second Semester	**12**	**2**	**13**	**20.5**

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Second Year Third Semester

Sl. No.	Category	Subject Code	Subject Name	Total Number of contact hours			Credits
				L	T	P	
Theory							
1	Basic Science course	BS-M301	Numerical Methods & Optimization	3	1	0	4
2	Basic Science course	BS-BIO301	Biology	3	0	0	3
3	Engineering Science Courses	ES-ROB301	Thermodynamics	3	0	0	3
4	Engineering Science Courses	ES-ME301	Engineering Mechanics	3	1	0	4
5	Professional Core courses	PC-ROB301	Analog & Digital Electronics	3	1	0	3
6	Professional Core courses	PC-ROB302	Manufacturing Processes	4	0	0	3
	Total Theory			**19**	**3**	**0**	**20**
Practical							
1	Professional Core courses	PC-ROB391	Analog & Digital Electronics Lab	0	0	3	1.5
2	Professional Core Courses	PC-ROB392	Practice of Manufacturing Processes	0	0	3	1
3	Professional Core courses	PC-ROB393	Machine Drawing Lab -I	0	0	3	1.5
	Total Practical			**0**	**0**	**9**	**4**
			Total of Third Semester	**19**	**3**	**9**	**24**

Second Year Fourth Semester

Sl. No.	Category	Subject Code	Subject Name	Total Number of contact hours			Credits
				L	T	P	
Theory							
1	Engineering Science Courses	ES-ROB401	Materials Engineering	3	0	0	3
2	Professional Core courses	PC-ROB 401	Fluid Power & Control	3	1	0	4
3	Professional Core courses	PC-ROB 402	Strength of Materials	3	1	0	4
4	Professional Core courses	PC-EE 403	Electrical & Electronics Measurements	3	1	0	3
5	Professional Core courses	PC-EE401	Electric Machines	3	1	0	3
	Total Theory			**15**	**4**	**0**	**17**
Practical/Sessional							
1	Professional Core courses	PC-ROB491	Robotics Laboratory I (Strength of Materials)	0	0	3	1.5
2	Mandatory courses	MC 481	Environmental Science	0	0	2	0
3	Professional Core courses	PC-EE491	Electric Machines Laboratory	0	0	2	1
4	Professional Core courses	PC-ROB491	Robotics Laboratory II (Fluid Power & Control)	0	0	3	1.5
5	Professional Core courses	PC-EE493	Electrical & Electronics Measurements lab	0	0	2	1
	Total Practical			**0**	**0**	**12**	**5**
			Total of Fourth Semester	**15**	**4**	**12**	**22**

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Third Year Fifth Semester

Sl No.	Category	Subject Code	Subject Name	Total Number of contact hours			Credits
				L	T	P	
Theory							
1	Professional Core courses	PC-ROB501	Design of Machine Elements	3	1	0	3
2	Professional Core courses	PC-EE501	Power Electronics	3	1	0	3
3	Professional Core courses	PC-ROB502	Kinematics & Theory of Machines	3	1	0	3
4	Humanities and Social Sciences including Management courses	HM-HU501	Humanities I	3	0	0	2
5	Professional Core courses	PC-EC501	Microprocessor & Microcontrollers	3	1	0	3
6	Mandatory courses	MC501	Essence of Indian Knowledge Tradition	2	-	-	0
	Total Theory			**17**	**4**	**0**	**14**
Practical/ Sessional							
1	Professional Core courses	PC-ROB591	Robotics Laboratory III- Practice of Manufacturing Processes and Systems Laboratory	0	0	3	1.5
2	Professional Core courses	PC-ROB592	Machine Drawing II	0	0	3	1.5
3	Professional Core courses	PC-ROB593	Microprocessor & Microcontrollers Lab	0	0	3	1
4	Professional Core courses	PC-EE591	Power Electronics Laboratory	0	0	2	1
5	Project (Summer internship)	PW-ROB581	Project-I (30 hrs. Total)	0	0	2	1
	Total Practical			**0**	**0**	**13**	**6**
			Total of Fifth Semester	**17**	**4**	**13**	**20**

Third Year Sixth Semester

Sl.No.	Category	Subject Code	Subject Name	Total Number of contact hours			Credits
				L	T	P	
Theory							
1	Professional Core courses	PC-CS601	Artificial Intelligence & Machine Learning	4	0	2	3
2	Professional Core courses	PC-ROB602	Computer Vision	3	1	0	3
3	Professional Elective courses	PE-ROB601A/B/C/D	Elective-I/ Mechatronic Systems/Digital Signal Processing /Human Computer Interaction/ Pattern Recognition	3	0	0	3
4	Professional Elective courses	PE-ROB602A/B/C/D	Elective-II/ Bio-Medical Electronics/ Material Handling/ 3D Printing and Design/ CAD/CAM	3	0	0	3

5	Humanities and Social Sciences including Management courses	HM-HU601	Humanities II (OR)	3	0	0	3
6	Mandatory courses	MC601	Constitution of India	2	-	-	0
	Total Theory			**18**	**1**	**0**	**15**
Practical/ Sessional							
1	Professional Core courses	PC-ROB691	Computer Vision Lab	0	0	3	1
2	Professional Core courses	PC- CS691	Artificial Intelligence & Machine Learning Laboratory	0	0	2	1
3	Project (Summer internship)	PW- ROB681	Project-II (90 hrs. Total)	0	0	4	2
	Total Practical			**0**	**0**	**9**	**4**
	Total of Sixth Semester			**18**	**1**	**9**	**19**

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Fourth Year Seventh Semester

Sl No.	Category	Subject Code	Subject Name	Total Number of contact hours			Credits
				L	T	P	
Theory							
1	Professional Core courses	PC- ROB701	Automation in Manufacturing	3	0	0	3
2	Professional Core courses	PC- ROB702	Embedded System	3	0	0	3
3	Professional Elective courses	PE- ROB701A/B/C/ D	Elective-III/ Speech and Natural Language Processing/ Neural Network and Fuzzy Logic Control/Wireless Sensor Networks/Industrial Automation & Control	3	0	0	3
4	Open Elective courses	OE-ROB 701A/B/C/D	Open Elective- I/ Industrial Pollution and Control / Entrepreneurship/ Object Oriented Programming/ Cyber Law & Intellectual Property Rights	3	0	0	3
5	Humanities and Social Sciences including Management courses	HM-HU701	Economics for Engineers	2	0	0	2
	Total Theory			**14**	**0**	**0**	**14**
Practical/ Sessional							
1	Professional Core courses	PE- ROB791	Robotics laboratory IV- (Advanced Manufacturing Techniques)	0	0	3	1.5
2	Project	PW- ROB781	Project-III	0	0	6	3
	Total Practical			**0**	**0**	**9**	**4.5**
			Total of Seventh Semester	**14**	**0**	**9**	**18.5**

Fourth Year Eighth Semester

Sl No.	Category	Subject Code	Subject Name	Total Number of contact hours			Credits
				L	T	P	
Theory							
1	Professional Elective courses	PE- ROB801A/B/C	Elective IV/ Multi-agent Intelligent Systems/Internet of Things(IoT)/ Cloud Computing	3	0	0	3
2	Professional Elective courses	PE- ROB802A/B/C	Elective V/ Mobile Communication and Networks/ Antennas and Propagation/ Fiber Optic Communication	3	0	0	3
3	Open Elective courses	OE- ROB801A/B/C	Open Elective-II/ Big Data Analysis/Robotics/ Web Technology	3	0	0	3
4	Open Elective courses	OE- ROB802A/B/C	Open Elective- III/ Micro-Electronics and VLSI Design/ Microwave Integrated Circuits/ Nano Electronics	3	0	0	3
	Total Theory			**12**	**0**	**0**	**12**

Practical/ Sessional							
1	Project	PW- ROB881	Project-IV	0	0	10	5
2	Professional Core courses	PW- ROB882	Comprehensive viva	0	0	0	1.5
	Total Practical			**0**	**0**	**10**	**6.5**
	Total of Eight Semester			**12**	**0**	**10**	**18.5**
	Total			117	17	84	**160**